Core Focus on Proportions & Probability

Block 3 Review ~ Percents

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

1. Keisha made a list of numbers she thinks are equal. Check her list. Circle all that apply.

A. 0.07 = 7%

B. 2.4 = 24%

C. 0.009 = 9%

D. $\frac{3}{4}$ = 75%

E. $\frac{2}{5}$ = 25%

F. $\frac{1}{3} = 33\frac{1}{3}\%$

A, D, F. (B: 2.4 = 240%. C: 0.009 = 0.9%. E: 2/5 = 40%.)

2. What is 20% of 75?

A. 1.5

B. 15

C. 25

D. 60

B. 0.2 × 75 = 15

3. Twelve is 4% of what number?

A. 0.48

B. 3

C. 300

D. 480

C. 12 = 0.04x → x = 12 ÷ 0.04 = 300

4. What percent of 25 is 40?

A. 15%

B. 60%

C. 62.5%

D. 160%

D. 40 ÷ 25 = 1.6 = 160%

5. What is the percent decrease from 50 to 35?

A. 15%

B. 30%

C. 70%

D. 130%

B. Change = 15; 15 ÷ 50 = 0.30 = 30%

6. Last year, 8 students joined the garden club. This year, 14 joined. What is the percent increase?

A. 6%

B. 42.9%

C. 75%

D. 175%

C. Change = 6; 6 ÷ 8 = 0.75 = 75%

7. Omar bought a pair of shoes for $40. His state has a 6% sales tax. How much money did Omar pay in sales tax?

A. $0.24

B. $2.40

C. $6.00

D. $24.00

B. 0.06 × 40 = $2.40

8. **Dev bought new headphones. The original price was $60, but they were on sale for 25% off. For numbers 8a – 8d, determine whether each statement about Dev's purchase is true or false.**

8a. Dev paid less than $60 for the headphones. **TRUE FALSE**

8b. The discount was $25. **TRUE FALSE**

8c. Dev paid $45 for the headphones. **TRUE FALSE**

8d. Dev paid $75 for the headphones. **TRUE FALSE**

8a TRUE · 8b FALSE (0.25 × 60 = $15) · 8c TRUE ($60 − $15 = $45) · 8d FALSE ($75 adds the discount)

Form B

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Part II – Constructed Response

Write each decimal or fraction as a percent.

1. 0.35

35% (0.35 × 100)

2. $\frac{2}{5}$

40% (2/5 = 40/100)

Solve. Show all work necessary to justify your answer.

3. What is 30% of 270?

81 ($x = 0.3 \times 270$)

4. Nine is 12% of what number?

75 ($9 = 0.12x \rightarrow x = 9 \div 0.12$)

5. What percent of 40 is 10?

25% ($10 = x \cdot 40 \rightarrow x = 0.25$)

6. Find the percent of decrease from 45 to 36.

20% decrease (change 9; 9 ÷ 45 = 0.2)

Find each selling price. Use mathematics to justify your answer.

7. original price: $48.00
percent of markup: 25%

$60.00 (markup 0.25 × 48 = $12; $48 + $12)

8. original price: $40.00
percent of discount: 15%

$34.00 (discount 0.15 × 40 = $6; $40 − $6)

9. A $240 tablet was on sale for 25% off the original price. How much money does a person save if he buys it on sale? Show all work necessary to justify your answer.

$60 saved (0.25 × 240 = 60)

10. The library book sale raised $800 last year. This year the library hopes to raise 15% more money. How much money does the library hope to raise this year? Show all work necessary to justify your answer.

$920 (0.15 × 800 = $120 more; $800 + $120 = $920)

11. Jamal bought a new TV. The TV cost $400. His city has an 8.5% sales tax. Jamal did the work shown below to figure out how much he would pay for the TV, including the sales tax. Show another way to find the total amount Jamal paid for the TV.

Let x = tax to pay

$x = 400 \cdot 0.085$
$x = 34$
I will pay $34.00 in tax.

Total cost for the TV:
$400 + $34.00 = $434.00

I will pay $434.00 for the TV.

$434.00. Sample: he pays 100% + 8.5% = 108.5% of the price, so 1.085 × 400 = $434.00. Or a proportion: $108.5/100 = x/400 \rightarrow x = 434$.

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Part III – Problem Solving

A winter coat was originally $120. It was on sale for 20% off the original price. When Elena got to the register, she found out it was an additional 25% off the sale price. What was the overall percent of discount from the original price of the coat to the price Elena paid?

Sale price: 0.8 × 120 = $96. Then 25% off: 0.75 × 96 = $72. Savings: $120 − $72 = $48. Overall discount: 48 ÷ 120 = 0.40 = **40%** (not 45%). Check: 0.8 × 0.75 = 0.6, so she paid 60% of the original.